



Innovation report

A: Static Structural

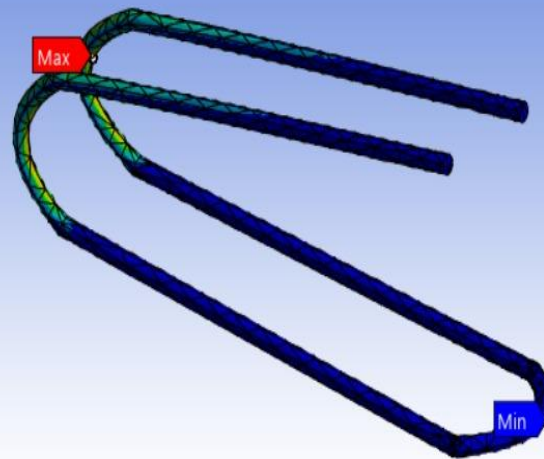
Equivalent Stress

Type: Equivalent (von-Mises) Stress

Unit: Pa

Time: 1 s

23-09-2025 12:07:57



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65Mn steel

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2

Model (A4) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	<i>Geometry Import (A3)</i>
State	Solved
Definition	
Source	C:\Users\bhuvanan\Downloads\SPRING.stp
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS; DS
Attributes	No

Named Selections	No
Material Properties	yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts on Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies on Import	No
Stitch Surfaces on Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\bhuvanan\Downloads\spring.stp
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	1.0727 m
Length Y	0.63462 m
Length Z	0.13 m
Properties (65Mn steel)	
Density	7850 kg/m3
Poisson's ratio	0.30
Young's modulus	210 Gpa
Yield strength	1000Mpa
Tensile strength	1200Mpa
Statistics	
Bodies	1
Active Bodies	1
Nodes	5000
Elements	2105
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No

Parameters	Independent
Parameter Key	ANS; DS
Attributes	No
ONamed Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts on Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies on Import	No
Stitch Surfaces on Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	CHASIS-FreeParts CHASIS
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	65Mn steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	1.0727 m
Length Y	0.63462 m
Length Z	0.13 m
Properties	
Volume	1.1039e-003 m ³
Mass	8.688 kg
Centroid X	0.49039 m
Centroid Y	-2.7451e-002 m
Centroid Z	-6.8743e-008 m
Moment of Inertia Ip1	0.11513 kg·m ²
Moment of Inertia Ip2	0.42826 kg·m ²
Moment of Inertia Ip3	0.53694 kg·m ²
Statistics	
Nodes	7965

Elements	3823
Mesh Metric	None

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	2
Material Assignments	1

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0
Origin	
Origin X	0 m
Origin Y	0 m
Origin Z	0 m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Mesh

TABLE 8
Model (A4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeathering	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly

Bounding Box Diagonal	1.2531 m
Average Surface Area	4.6191e-003 m²
Minimum Edge Length	1.4211e-003 m
Quality	
Check Mesh Quality	Yes
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No

TABLE 9
Model (A4) > Mesh > Mesh Controls

Object Name	<i>Face Meshing</i>
State	Pass
Scope	
Scoping Method	Geometry Selection
Geometry	8 Faces
Definition	
Suppressed	No
Mapped Mesh	Yes
Constrain Boundary	No
Advanced	
Specified Sides	No Selection
Specified Corners	No Selection
Specified Ends	No Selection
Multizone Semi-Structured	No

Static Structural (A5)

TABLE 10
Model (A4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 11
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1
Current Step Number	1
Step End Time	1s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotor dynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled

Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\bhuvanan\AppData\Local\Temp\WB_saroj_26664_2\wbnew_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 12
Model (A4) > Static Structural (A5) > Loads

Object Name	<i>Fixed Support</i>	<i>Fixed Support 2</i>	<i>Force</i>
State	Fully Defined		

Scope			
Scoping Method	Geometry Selection		
Geometry	6 Faces	2 Faces	1 Face
Definition			
Type	Fixed Support		Force
Suppressed	No		
Define By			Components
Applied By			Surface Effect
Coordinate System			Global Coordinate System
X Component			0. N (ramped)
Y Component			-490.5 N (ramped)
Z Component			0. N (ramped)

Solution (A6)

TABLE 13
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1
Refinement Depth	2
Information	
Status	Done
MAPDL Elapsed Time	3s
MAPDL Memory Used	562 MB
MAPDL Result File Size	2.75 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

FIGURE 1
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Total Deformation

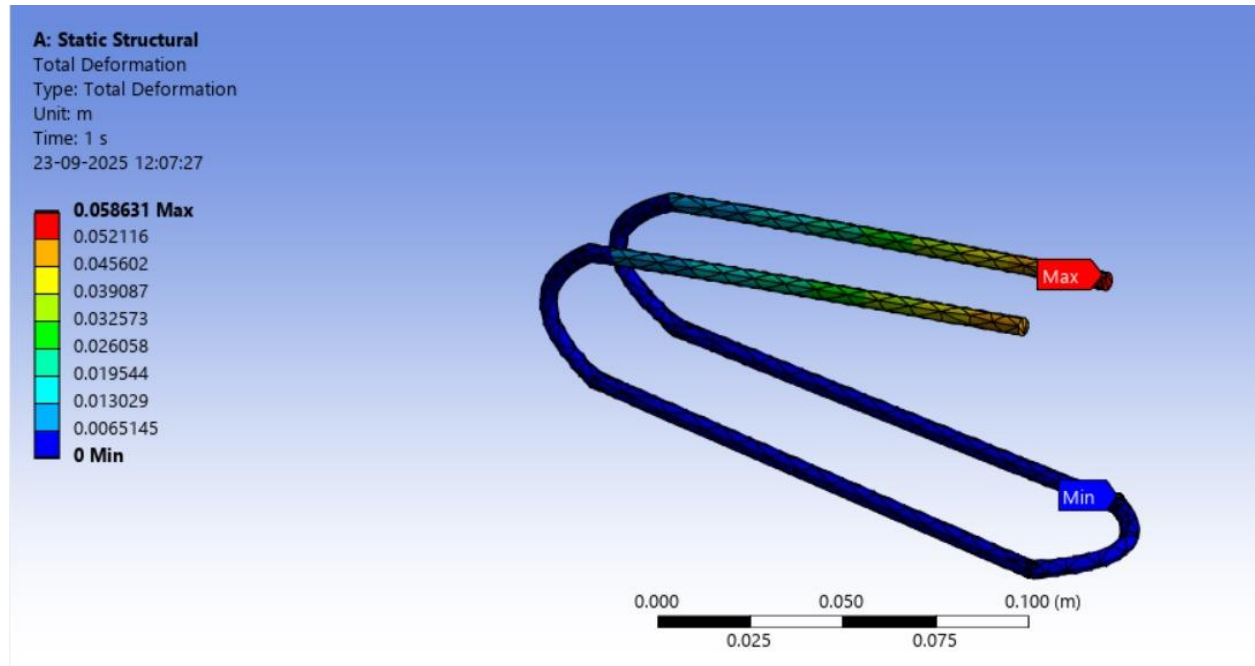


TABLE 16

Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation

NEWTON	N1 [680]	N2 [1360]	N 3 [2040]
DEFORMATION	0.05863M	0.09587M	0.10058 M

FIGURE 7
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Equivalent stress

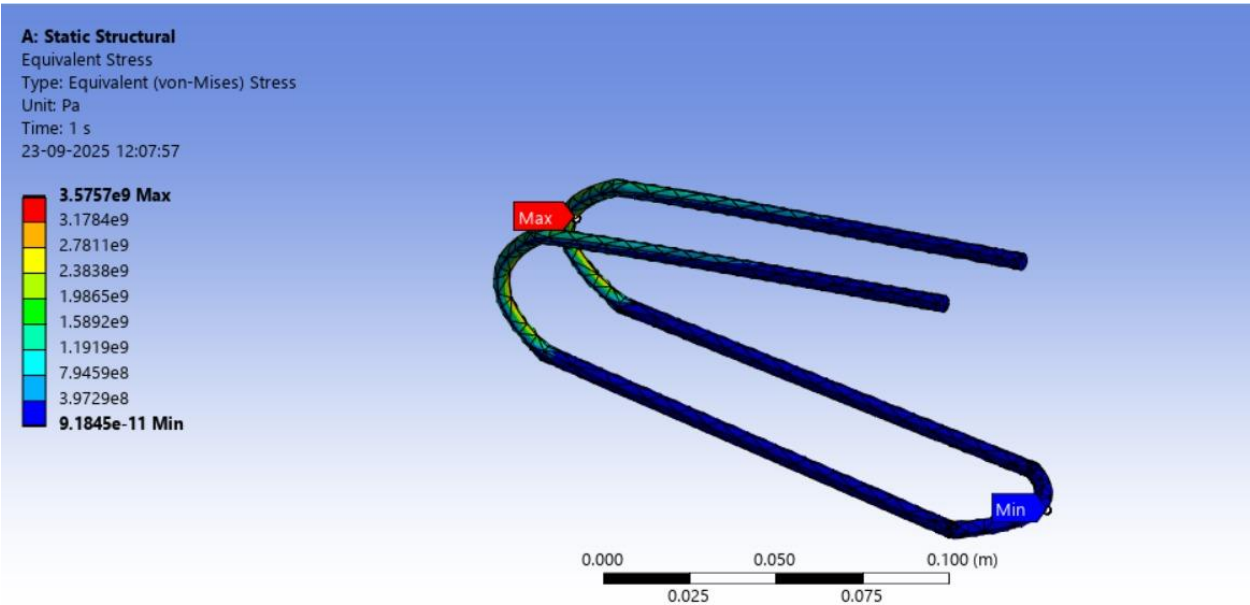


TABLE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

NEWTON	N1[680]	N2 [1360]	N3 [2040]
EQUIVALENT STRESS	3.5757pa	6.3801pa	8.0702pa

FIGURE 9
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain > Equivalent strain

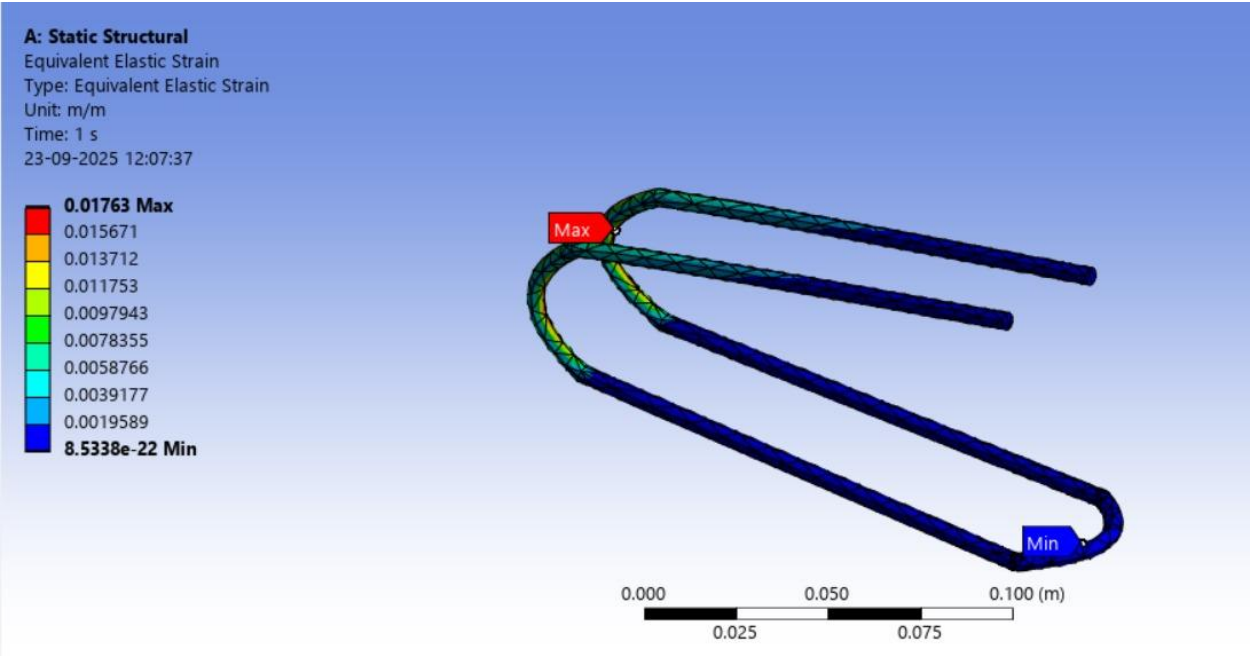


TABLE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain

NEWTON	N1 [680]	N 2 [1360]	N3 [2040]
ELASTIC STRAIN	0.01763m/m	0.0246m/m	0.0387m/m

Material Data

65Mn steel

TABLE 20
65Mn steel > Constants

Density	7850 kg m ⁻³
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TABLE 22
65Mn steel > Isotropic Elasticity

Young's Modulus GPa	Poisson's Ratio	Bulk Modulus GPa	Shear Modulus GPa
210	0.30	175	80.8

TABLE 23
65Mn steel > Tensile Yield Strength

Tensile Yield Strength MPa
900

TABLE 24
65Mn steel > Tensile Ultimate Strength

Tensile Ultimate Strength Mpa
1200

